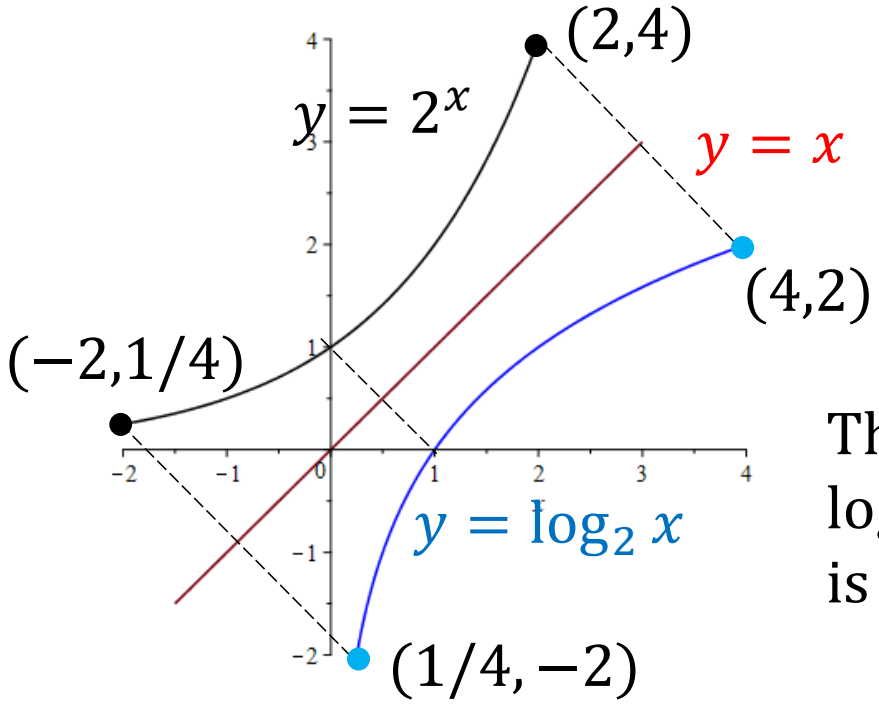
2.5 Logarithmic Functions

We note that, for $a > 1$, the graph G of a^x is rising in $(-\infty, \infty)$. If we reflect the graph G about the line $y = x$, we obtain a curve G' which is rising in $(0, \infty)$, and hence G' is a graph of an increasing function. The function corresponding to graph G' is called the *logarithmic function with base* a, denoted by $\log_a x$.



Exponential function
and
Logarithmic function

The domain of a
logarithmic function
is $(0, \infty)$

Inverse Relation Between Exponential and Logarithmic Functions

From the definition, we note the following relations. The point (c, d) lies in the graph of $\log_a x$ if and only if the point (d, c) lies in the graph of a^x, that is,

$$d = \log_a c \iff c = a^d$$

The exponential function and the logarithmic function *undo each other*, that is,

$$a^{\log_a c} = c, \;\; \log_a a^d = d.$$

They form a pair of *inverse functions*.

2.5 Laws of Logarithms

The domain of an exponential function is $(-\infty, \infty)$, and the domain of a logarithmic function is $(0, \infty)$. Let a, b, x and y be real numbers with $a > 0,\ b > 0$. Then

$$\log_a 1 = 0, \qquad a^0 = 1,$$

$$\log_a(xy) = \log_a x + \log_a y, \qquad a^{x+y} = a^x a^y,$$

$$\log_a(x/y) = \log_a x - \log_a y, \qquad a^{x-y} = \frac{a^x}{a^y},$$

$$\log_a(x^y) = y \log_a x, \qquad (a^x)^y = a^{xy}.$$

2.5 Laws of Logarithms

Example We have

$$\log_3 9 = \log_3 3^2 = 2,$$
$$\log_2(1/8) = \log_2 1 - \log_2 8 = 0 - \log_2 2^3 = -3$$
$$\log_2(1/\sqrt{8}) = \log_2 8^{-1/2} = -\frac{1}{2}\log_2 8 = -\frac{3}{2}.$$

Example Solve the equation $3^{2x} \times 2^x = 324$.
Solution We first rewrite the equation as

$$\left(3^2\right)^x \times 2^x = 18^2 \Longleftrightarrow 18^x = 18^2.$$

Hence $x = 2$.

2.5 The Natural Logarithmic Function

The function $\log_e x$ is called the *natural logarithmic function*, and it is denoted by $\ln x$. Thus

$$y = \ln x \iff x > 0 \text{ and } e^y = x.$$

Change of Base Formula Let a, b, c be positive numbers. Then

$$\log_a b = \frac{\log_c b}{\log_c a}.$$

In particular,

$$\log_a b = \frac{\ln b}{\ln a}.$$

We also have

$$a^x = \left(e^{\ln a}\right)^x = e^{(\ln a)x}.$$

2.5 Inverse Function

Example Express x^x as a composite function of exponential and logarithmic functions with base e.
Solution We have

$$x^x = \left(e^{\ln x}\right)^x = e^{x \ln x}.$$

If we let $f(x) = e^x$ and $g(x) = x \ln x$, then

$$x^x = f(g(x)).$$

2.5 Inverse Function

Consider a function $f(x)$ and its graph G. Let G' be obtained by reflecting G about the line $y = x$. Suppose G passes the *Horizontal Line Test*: each horizontal line intersects G at most once.
Then each vertical line intersects G' at most once, which implies that G' is the graph of some function, which is called the *inverse function* of f, denoted by $f^{-1}(x)$.
We note that the the graph of f passes Horizontal Line Test means that different input of f gives different output. That is,

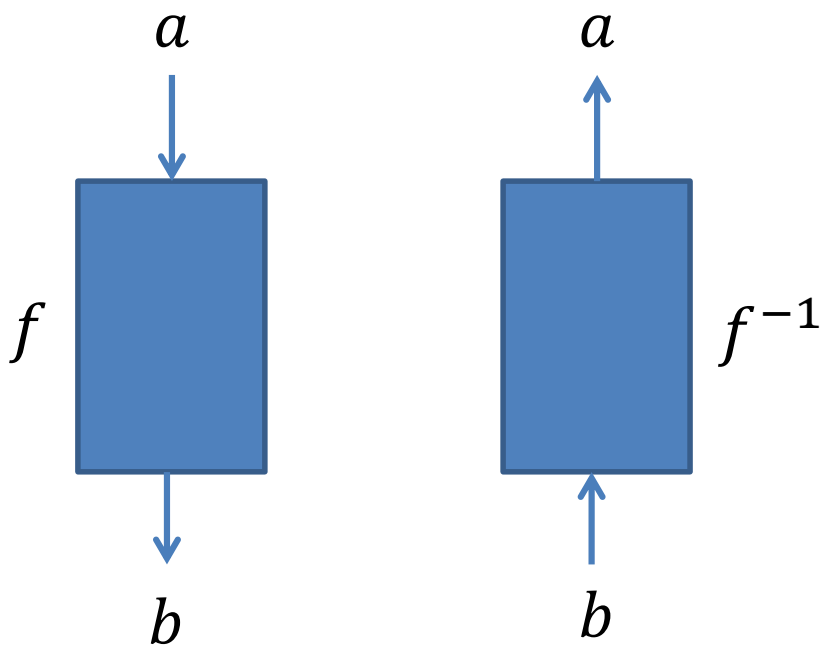
$$x_1 \neq x_2 \Longrightarrow f(x_1) \neq f(x_2). \tag{1}$$

2.5 Inverse Function

Fact $f(a) = b \iff f^{-1}(b) = a.$

$$f\left(f^{-1}(x)\right) = f^{-1}(f(x)) = x,$$

$$Dom\{f\} = Range\{f^{-1}\}, Range\{f\} = Dom\{f^{-1}\}.$$



$$f(a) = b \iff f^{-1}(b) = a$$

2.5 Inverse Function

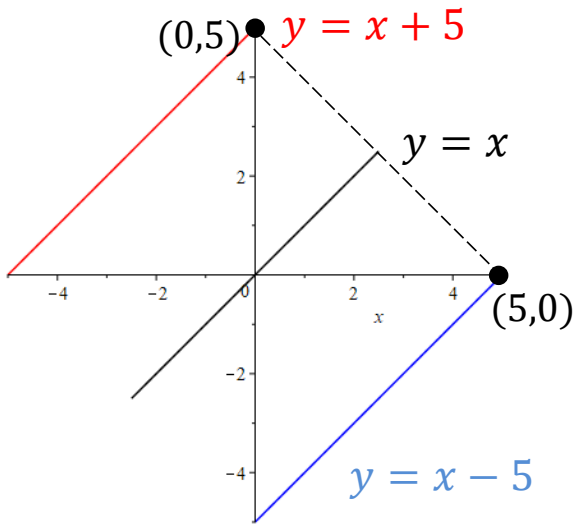
A function f satisfies property (1) is called a 1-to-1 function.

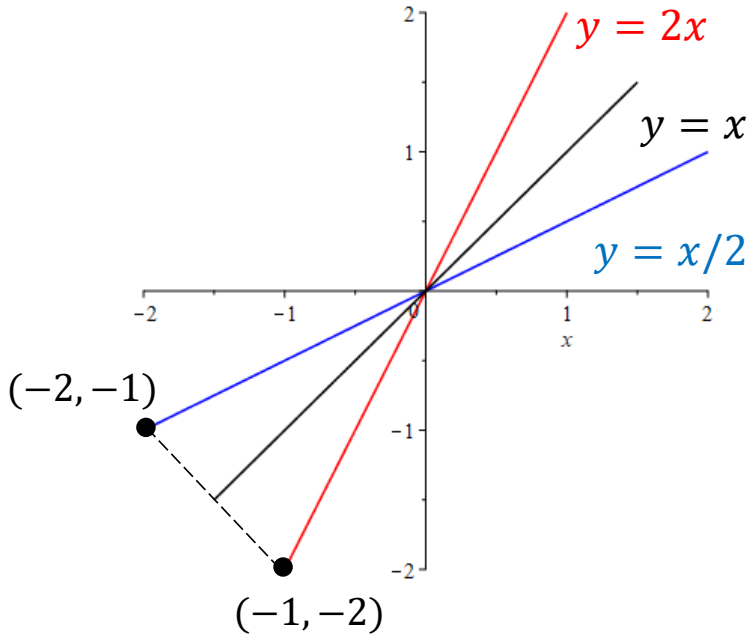
Example For any constant $a > 1$, the exponential function a^x with domain $(-\infty, \infty)$ is 1-to-1. Hence it has an inverse function, denoted by $\log_a x$. We have $Range\{a^x\} = (0, \infty)$, $Dom\{\log_a x\} = (0, \infty)$, $Range\{\log_a x\} = (-\infty, \infty)$.

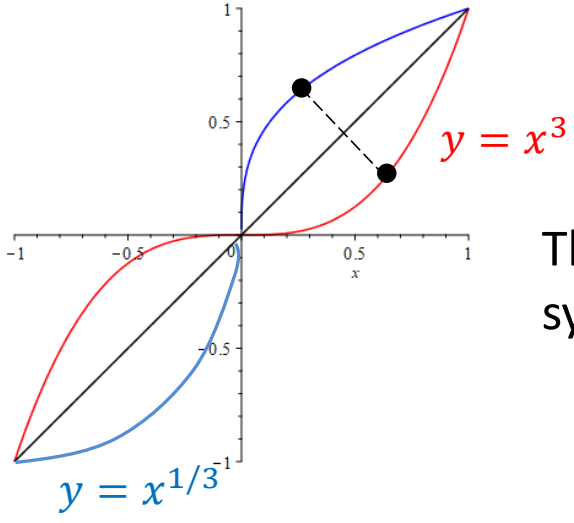
Example The function x^3 with domain $(-\infty, \infty)$ is 1-to-1. Its inverse function is $x^{1/3}$ (or $\sqrt[3]{x}$). We have $Range\{x^3\} = (-\infty, \infty)$, $Dom\{x^{1/3}\} = (-\infty, \infty)$, $Range\{x^{1/3}\} = (-\infty, \infty)$.

Inverse function

Examples of pair of inverse functions.







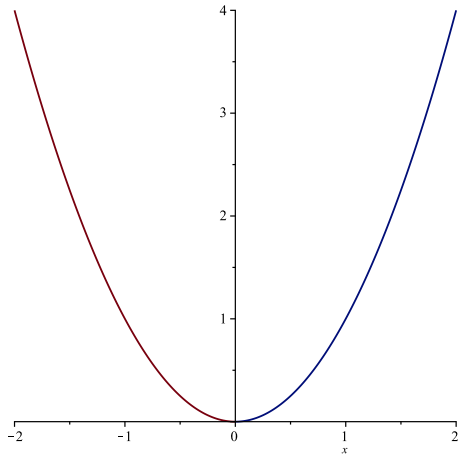
The graphs of inverse functions are symmetric about the line $y = x$

2.5 Inverse Function

Example The function $f(x) = x^2$ with domain $(-\infty, \infty)$ is not 1-to-1, and so it does not have an inverse function.

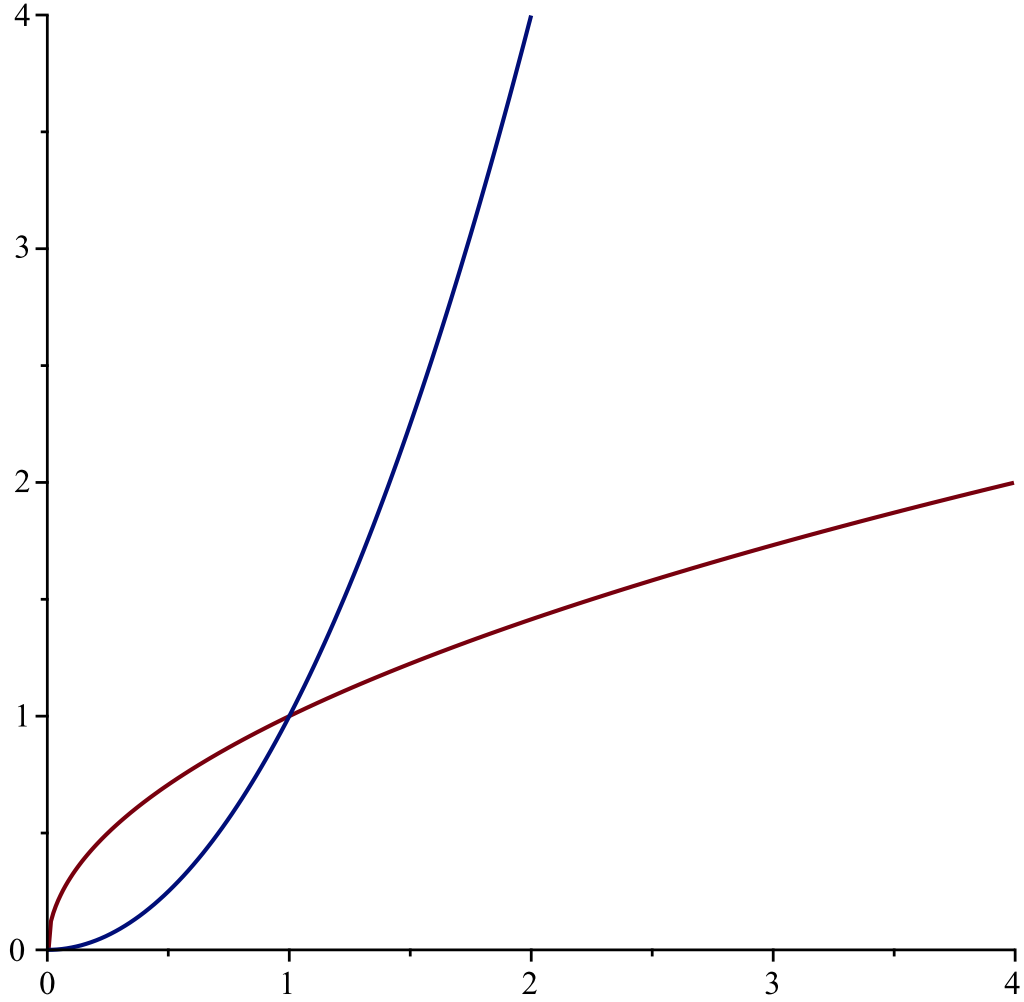
Example The function $g(x) = x^2$ with domain $[0, \infty)$ is 1-to-1, and it has an inverse function, which is $x^{1/2}$ with domain $[0, \infty)$ and range $[0, \infty)$.

Example The function $h(x) = x^2$ with domain $(-\infty, 0]$ is 1-to-1 and its inverse function $-x^{1/2}$.



Inverse Function

The function x^2 with domain $[0, \infty)$ and its inverse function $g(x) = x^{1/2}$.



2.6 Simple interest

Example Suppose \$1000 is invested at the nominal rate of 6% (annual interest rate). Then the accumulated amount after one year is

$$1000 + 1000 \times 0.06 = 1000 \times 1.06.$$

Suppose the interest does not earn interest (not compounding), then the accumulated amount after two years will be

$$1000 + 1000 \times 0.06 \times 2 = 1000(1 + 2 \times 0.06).$$

The accumulated amount after t years will be

$$1000 + 1000 \times 0.06 \times t = 1000(1 + 0.06t).$$

2.6 Simple interest

In general, let P be the principal amount, and r be the nominal rate (annual interest rate). Then the *simple interest* after t years is

$$I(t) = Prt.$$

The accumulated amount after t years (with simple interest) will be

$$A(t) = P + Prt = P(1 + rt).$$

This is a linear function of t.

2.6 Compound interest

Example Suppose \$1000 is invested at the nominal rate of 6%, and the interest is compounded (earning interest). Then the accumulated amount after one year is

$$1000 + 1000 \times 0.06 = 1000 \times 1.06.$$

After two years the accumulated amount will be

$$1000 \times 1.06 + 1000 \times 1.06 \times 0.06 = 1000 \times 1.06^2.$$

After t years the accumulated amount will be

$$1000 \times 1.06^t,$$

which is an exponential function of t.

2.6 Compound interest

Example Suppose the nominal rate is 0.06. How long does it take for the principal amount to be doubled with
(a) simple interest, (b) compound interest?
Solution (a) We solve the equation
$P(1+0.06t)=2P$ to obtain
$0.06t=1 \Longrightarrow t \approx 16.7$ (years)
(b) We solve the equation $P \times 1.06^t = 2P$ to obtain

$$1.06^t = 2, \;\; \ln(1.06^t) = \ln 2, \;\; t \ln 1.06 = \ln 2,$$

$$t = \frac{\ln 2}{\ln 1.06} \approx 11.9 \text{ (years)}.$$